

Quantification: 682_z23

=====

PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image

measure	value
-----	-----
lacuna count	14
interior lacuna count	11
frame-edge lacuna count	3
gap bridges added (detection)	19

Interior lacunae only: mean, median and sample SD (n = 11)

measure	kind	mean	median	sd	units
-----	-----	-----	-----	-----	-----
area_px2	lacuna shape	1994.91	1519	1026.72	px^2
major_axis_length_px	lacuna shape	85.7455	86.1566	22.5145	px
minor_axis_length_px	lacuna shape	30.4546	26.8577	8.8339	px
aspect_ratio	lacuna shape	2.9668	3.0363	0.8704	unitless
eccentricity	lacuna shape	0.92	0.9442	0.0643	unitless
solidity	lacuna shape	0.8555	0.8762	0.0782	unitless
roots_count	headline	6.7273	6	1.9022	count
ring_length_r30_px	headline	283.364	306	68.4621	px
ring_length_r60_px	reported	795.182	751	131.757	px
owned_length_px	ownership-dependent	1065.82	920.896	819.239	px
edge_count	ownership-dependent	35	31	27.5971	count
mean_edge_length_px	ownership-dependent	32.7335	31.7551	7.1512	px
perimeter_px	normalised	221.572	210.244	51.1926	px
ring_area_r30_px2	normalised	8909.82	8646	1432.67	px^2
ring_area_r60_px2	normalised	22488.8	22056	3166.3	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9972	1	0.0094	unitless
ring_density_r30	normalised	0.0320181	0.0306724	0.00806864	px^-1
ring_density_r60	normalised	0.0356529	0.0343229	0.00603288	px^-1
roots_per_100px_perimeter	normalised	3.0793	2.8652	0.7844	per 100 px
ring_attached_length_r30_px	network v2	213.273	215	77.2284	px
ring_attached_length_r60_px	network v2	419.454	444	153.073	px
ring_length_w_r30_px	network v2	315.341	341.35	74.1913	px
ring_length_w_r60_px	network v2	890.997	845.465	145.162	px
sholl_crossings_r10	network v2	7.4545	7	2.0671	count
sholl_crossings_r20	network v2	10.8182	10	2.316	count
sholl_crossings_r30	network v2	13	13	2.8284	count
ring_width_mean_r30_px	width	5.2823	5.2403	0.3396	px

Per field

measure	value	units
-----	-----	-----
analysed_area_px2	1.02271e+06	px^2
total_skeleton_length_px	39168	px
canalicular_length_density_per_px	0.0382982	px^-1
junction_count	481	count
junction_density_per_px2	0.00047032	px^-2
lacuna_count	14	count
lacunae_per_px2	1.369e-05	px^-2

skeleton_component_count	756	count
mean_component_length_px	51.8095	px
median_component_length_px	13.5	px
field_length_density_w_per_px	0.0430341	px^-1
field_density_without_flagged_per_px	0.0380563	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	5.8663	px
width_median_px	6	px
width_p10_px	4	px
width_p90_px	8	px
width_pixels_used	36498	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels with real signal under them: bridged pixels, the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	510	36.0388	20.5481	1.7539	0.8215	0.8095	-0.2657	13.3863	106.873	True
2	448	43.7126	19.8458	2.2026	0.891	0.6767	1.435	7.0045	363.067	True
3	1379	89.4664	21.3431	4.1918	0.9711	0.7971	-1.5436	60.8564	471.593	False
4	1062	83.8665	19.2697	4.3522	0.9732	0.6923	-1.1353	135.124	920.395	False
5	4132	136.42	39.4244	3.4603	0.9573	0.9008	1.5344	431.979	214.138	False
6	1032	65.9485	21.7201	3.0363	0.9442	0.8418	-0.9507	441.636	703.582	False
7	2868	91.0398	40.6809	2.2379	0.8946	0.9166	-1.49	534.719	477.113	False
8	980	60.8458	24.7024	2.4632	0.9139	0.7644	-1.4596	557.819	348.526	False
9	2962	111.965	35.5084	3.1532	0.9484	0.917	1.4275	591.514	47.6516	True
10	2810	108.692	33.8251	3.2134	0.9503	0.9201	1.4475	714.117	172.09	False
11	1297	66.0375	26.8577	2.4588	0.9136	0.9134	-1.189	725.743	851.213	False
12	1519	86.1566	25.1812	3.4215	0.9563	0.8388	1.3918	837.228	113.164	False
13	2083	63.4325	42.3049	1.4994	0.7451	0.9485	-1.2565	905.171	752.206	False
14	2782	91.2948	39.6911	2.3001	0.9005	0.8762	-1.4405	919.61	300.233	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	136	378	297.066	15	19.8044	
2	2	41	171	16.2426	2	8.1213	
3	5	294	711	278.995	9	30.9994	5.6388
4	8	328	881	1121.43	42	26.7007	5.1717
5	9	337	881	1245.97	53	23.509	5.4973
6	5	207	751	216.108	6	36.0179	5.2403
7	8	306	746	1597.01	50	31.9402	5.136
8	8	380	850	3058.9	100	30.589	5.7122
9	6	260	666	1034.67	35	29.5619	5.4807
10	6	312	1048	920.896	29	31.7551	5.6867
11	5	194	565	486.031	13	37.387	5.1124
12	5	256	741	444.446	9	49.3829	4.5534
13	5	166	676	781.938	31	25.2238	5.2621

14 10 337 897 1572.25 43 36.5639 5.0941

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	105.154	3701	10161	0.6475	0.597	0.0367468	0.0372011	3.8039
2	132.847	3699	7686	0.6031	0.5656	0.0110841	0.0222482	1.5055
3	219.681	8976	20715	1	0.9689	0.032754	0.0343229	2.276
4	210.244	8511	22591	1	1	0.0385384	0.0389978	3.8051
5	335.421	12178	29932	1	1	0.0276728	0.0294334	2.6832
6	167.238	7485	20582	1	1	0.0276553	0.0364882	2.9898
7	243.522	9472	22056	1	1	0.0323057	0.033823	3.2851
8	167.924	7349	17555	1	1	0.0517077	0.0484192	4.7641
9	282.894	8678	19332	0.813	0.7462	0.0299608	0.0344506	2.1209
10	263.48	10172	24552	1	1	0.0306724	0.0426849	2.2772
11	174.51	7570	20770	1	1	0.0256275	0.0272027	2.8652
12	210.167	8646	22189	1	1	0.0296091	0.0333949	2.3791
13	185.823	7980	21610	1	1	0.020802	0.0312818	2.6907
14	259.279	9669	24825	1	1	0.0348537	0.0361329	3.8568

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	118	260	148.569	409.907	3	5	5
2	28	28	39.3137	170.498	1	0	1
3	108	220	312.066	768.772	6	10	15
4	274	562	374.161	1004	10	10	13
5	246	444	367.836	971.566	9	14	11
6	133	163	227.48	845.465	5	9	9
7	262	447	341.35	828.512	8	10	12
8	327	610	417.563	934.997	9	13	13
9	181	338	288.208	743.884	7	10	11
10	191	491	348.421	1173.69	6	14	18
11	144	297	216.581	647.541	6	9	11
12	215	400	295.978	833.281	7	10	14
13	132	348	192.167	782.495	5	7	10
14	314	632	375.149	1010.65	11	13	17

